OMAR EZ-ELDIN ABDULLAH

omarfayed241@gmail.com | (+20) 01556595823 | Cairo, Egypt | LinkedIn | GitHub | Codeforces | Portfolio

PROFILE

AI Engineer and Computer Science graduate from Cairo University (FCAI) specializing in Agentic AI and Applied Deep Learning. Experienced in engineering end-to-end RAG architectures, local-first LLM agents, and automated computer vision pipelines. Proficient in PyTorch and LangGraph, with a strong focus on data engineering, domain adaptation, and deploying optimized local inference systems for highly accurate, low-latency AI applications.

EDUCATION

B.Sc in Computer Science **10/2022 – 07/2026**

Faculty of Computers and Artificial Intelligence, Cairo University *Cairo, Egypt*

- **GPA:** 3.3/4.0

WORK EXPERIENCE

Software Engineering Intern **07/2025 – 10/2025**

Microsystems Cloud Business Solution (*MCBS*) *Cairo, Egypt*

- **Frontend & Web Development:** Developed modular web applications and integration workflows using TypeScript, React, and Angular, managing build pipelines (Vite, Webpack) and E2E testing to ensure system reliability.
- **Network & Endpoint Security:** Conducted system security assessments by identifying open ports, discovering endpoint vulnerabilities, and executing penetration testing to document mitigation strategies.

AI Trainer / LLM Evaluator **05/2024 – 01/2026**

Scale AI *San Francisco, CA* (*remote*)

- Evaluated, annotated, and refined LLM outputs across **250+** production tasks spanning diverse projects with distinct technical requirements.
- Authored detailed justifications and validated model outputs across OOP, frontend architecture, computer vision, calculus, algebra, and bilingual (Arabic/English) data, partnered with QMs to resolve edge cases and enforce strict quality standards.

SKILLS

Machine Learning & Computer Vision:
PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, ONNX, rembg U²-Net, Faster R-CNN, Wav2Vec2, Roboflow Inference.

Data Science & Analysis:
NumPy, Pandas, Matplotlib, Seaborn

Programming Languages:
Python, C++, SQL, Java, Dart, JavaScript/TypeScript.

LLM & MLOps:
LangChain, LangGraph, LangSmith, ChromaDB, Sentence Transformers, BM25 Hybrid Retrieval, Hugging Face, Ollama, llama.cpp, MLflow, Docker, AWS.

Backend & Core Engineering
System Design, REST APIs, PostgreSQL, Object-Oriented Programming (OOP), Data Structures & Algorithms.

PROJECTS

Seed Bank (Grad Project) Seed Quality Classification **10/2025 – 07/2026**

PyTorch, YOLOv8, Faster R-CNN, EfficientNet-B2, U2-Net, Synthetic Data Generation

- Built computer vision platform to detect, count, and classify individual seed quality/defects from top-down images.
- Engineered dual pipelines: **YOLOv8** single-stage for real-time inference; modified **Faster R-CNN** + **EfficientNet-B2** two-stage for high-accuracy defect classification.
- Integrated **U2-Net** segmentation on **Faster R-CNN** crops to remove background noise and isolate seed contours; achieved **97.47% accuracy, 96.89% F1** on maize defect classification.
- Built **MultiSeedGen** synthetic data engine to overcome bias/scarcity.

- Applied domain randomization, clutter, occlusion, shadows, camera noise, and scaling to close synthetic-to-real gap and improve real-world generalization.
- Established model registry/benchmarking workflow to version, evaluate, and archive candidates against fixed baselines before production promotion.

Local RAG AI Agent for Research
LangGraph, Ollama, Qwen3.5-4B, ChromaDB, BM25, BGE Embeddings, Sentence-Transformers, Streamlit, Tavily

- **Local-First Research Agent:** Engineered a cited research agent over an O'Reilly AI books using local **Qwen 3.5-4B** (Ollama), with optional Tavily web search for out-of-corpus data.
- **Agentic Retrieval Architecture:** Architected a LangGraph workflow (Fast/Deep profiles) featuring query expansion, hybrid BM25 + ChromaDB retrieval, LLM relevance grading, adaptive rewrites (step-back, fusion), and cited answer synthesis.
- **Efficient Offline Inference:** Optimized local execution via CPU-based BGE embeddings, persistent Chroma indexes, reciprocal-rank fusion, batched generation, model keep-alive, and Streamlit caching to minimize latency.
- **Evaluation & Impact:** Evaluated via Hit@K, Recall@K, and MRR; achieved 100% Hit@6 / Recall@6 across 20 queries (0.95 MRR). Local Qwen 3.5-4B sustained ~40 tokens/sec at an 8K context window.

Advanced RAG Architectures
LangChain, ChromaDB, Hugging Face (BGE v1.5), LangChain-Groq.

- **Query Processing:** Implemented translation (Multi-query, RAG-Fusion, Step-back, HyDE), construction (Text-to-SQL/Cypher), and semantic/logical routing.
- **Indexing & Retrieval:** Engineered chunk optimization, multi-representation (RAPTOR), specialized embeddings (ColBERT), and retrieval refinement (Re-ranking, CRAG, Self-RAG).

Voice Gender Classification Microservice
Wav2Vec 2.0, Scikit-learn, FastAPI, MLflow, Docker, AWS EC2

- **Architecture:** Built an end-to-end speech classification pipeline using a frozen Wav2Vec 2.0 backbone (Layer 6) and Logistic Regression to extract embeddings and classify voice gender directly from raw audio.
- **Evaluation & Results:** Achieved 100% in-distribution validation accuracy and **99.2% out-of-distribution (OOD) accuracy** on the Google FLEURS dataset.
- **Audio Preprocessing:** Built an in-memory preprocessing pipeline (16kHz resampling, mono conversion, duration checks) with format fallbacks to reliably decode diverse audio files (16/24/32-bit WAV).
- **Serving & Containerization:** Built an asynchronous **FastAPI** backend delivering **~45ms** CPU inference, tracked training experiments in **MLflow**, and containerized with Docker for reproducible **AWS** EC2 deployment.

Rock Loyalty System
Angular 18, TypeScript, RxJS, JWT, OAuth2

Developed the Angular 18 frontend for a multi-tenant restaurant loyalty platform. Built a customer portal enabling users to authenticate, view their points balance, and exchange points for discount vouchers, alongside an admin panel for restaurant management.

- Implemented a secure **JWT + refresh-token** auth flow access tokens stored in-memory, refresh tokens in HTTP-only cookies, with proactive silent refresh and automatic logout on expiry.
- Built **Google OAuth2 social login** and a full email-confirmation / forgot-password flow.
- Designed a **multi-tenant routing architecture** (/:restaurantId) allowing one deployed build to serve multiple restaurant clients.
- Created **functional HTTP interceptors** for automatic Bearer token injection and API key authentication.
- Applied **4 route guards** (authGuard, notAuthGuard, phoneGuard, notPhoneGuard) for fine-grained access control.
- Used **RxJS BehaviorSubject** as a reactive state store for user session management across the app.

CERTIFICATES

2023 ECPC Qualifications Day 5 | MCBS internship completion | Digitera Certificate

LANGUAGES

Arabic	English	French
Native	Proficient	Basic